GlobalRetail 360

Batch Retail Data Engineering Platform

SQL Server | Fabric Data Factory | Snowflake | Databricks | Power BI

Domain	Retail: omnichannel sales, inventory, customer and product analytics (batch)
Role of project	Portfolio project for entry-level Data Engineer / Analytics Engineer roles
Core technologies	SQL Server, Microsoft Fabric Data Factory, Snowflake, Databricks (PySpark, Delta Lake), Power BI, pytest
Key patterns	Medallion architecture, SCD Type 2, incremental loads, data quality, audit logging
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1. Executive Summary

GlobalRetail 360 is a production-style batch data platform for a multi-country retailer selling through stores and e-commerce. Customer and store data live in an on-premises SQL Server; product data arrives as ERP JSON exports; sales as POS and e-commerce Parquet files; inventory adjustments as periodic JSON event files.

The platform consolidates these into one trusted analytical layer. **Microsoft Fabric Data Factory** orchestrates ingestion and incremental loads. **Snowflake** serves as the raw landing and the final serving layer. **Databricks** runs the PySpark transformations that build Bronze, Silver and Gold Delta tables. **Power BI** delivers executive and operational dashboards.

The source data is deliberately dirty (duplicates, invalid emails, inconsistent codes, negative stock), so the pipeline demonstrates validation, quarantine of rejected records, dimensional modeling, history tracking and automated testing rather than just moving clean data.

2. Business Objectives

- Track historical changes in customer and product attributes (SCD Type 2).
- Standardize channel, currency and country codes across all sources.
- Build reusable conformed dimensions and fact tables.
- Ingest incrementally using watermarks instead of full reloads.
- Publish curated marts to Snowflake for reporting and build Power BI dashboards.
- Add data-quality checks, audit logging and unit tests so the pipeline is trustworthy and debuggable.

3. Architecture

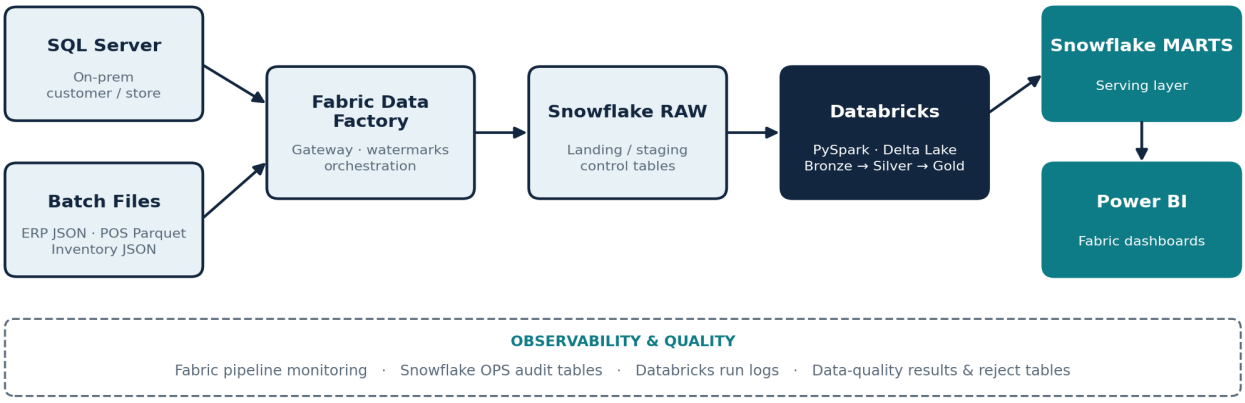


Figure 1. End-to-end flow from sources to dashboards.

Flow: SQL Server and batch files are ingested by Fabric Data Factory into Snowflake RAW. A Databricks job reads RAW, builds the medallion layers in Delta, and publishes Gold marts to Snowflake MARTS. Power BI in Fabric reads the marts. Snowflake acts as the exchange layer between Fabric and Databricks, so no cloud object storage account is required.

Scope: batch only. Inventory events are delivered as scheduled files, and low-stock monitoring is produced as batch exception reporting rather than real-time alerts.

4. Source Systems

Dataset	Origin / format	Data issues to handle
Customer master	On-prem SQL Server, incremental table extract	Duplicate business keys, invalid email, country-code variants, attribute changes over time
Store master	On-prem SQL Server, incremental table extract	Location normalization
Product catalog	ERP export, JSON	Category casing, invalid currency, price and attribute changes
Sales transactions	POS + e-commerce, Parquet	Duplicate order lines, zero quantity, missing product, mixed channel values
Inventory events	Operational export, batch JSON	Duplicate events, null keys, negative on-hand quantity

5. Ingestion and Orchestration

- Install an on-premises data gateway and create a SQL Server connection in Fabric Data Factory.
- Extract customer and store changes incrementally using a watermark column (*updated_at* / *modified_at*).
- Copy SQL Server data and the JSON / Parquet files into Snowflake RAW tables.
- Record pipeline and activity runs in Snowflake OPS audit tables.
- Trigger the Databricks transformation job only after ingestion succeeds.

6. Medallion Architecture

Layer	Purpose	Key objects
Bronze	Append-only raw history as Delta tables	customers_raw, stores_raw, products_raw, transactions_raw, inventory_events_raw
Silver	Validated, conformed analytical entities	dim_customer, dim_product, dim_store, dim_date, fact_sales, fact_inventory_event
Gold	Business-ready consumption marts	customer_analytics, inventory_alerts, sales_executive_mart
Ops	Audit, watermarks and data quality	watermark_config, notebook_run_log, data_quality_results, reject tables

7. Dimensional Model

Dimensions

Dimension	Pattern	Natural key	Surrogate key	Tracked attributes
dim_customer	SCD Type 2	customer_id	customer_sk	full_name, email, city, country_code, loyalty_tier, marketing_opt_in
dim_product	SCD Type 2	product_id	product_sk	product_name, category, brand, unit_price, unit_cost, reorder_point
dim_store	Type 1	store_id	store_sk	store_name, city, country_code, region, timezone_name
dim_date	Static	calendar_date	date_key	Standard reporting attributes

Facts

Fact	Grain	Measures	Foreign keys
fact_sales	order_id + order_line_id	quantity, gross_amount, net_amount, usd_net_amount, gross_margin_amount	customer_sk, product_sk, store_sk, date_key
fact_inventory_event	event_id	delta_qty, on_hand_after	product_sk, store_sk, date_key

8. SCD Type 2 Design

- Deduplicate to the latest source record per natural key using *updated_at* and ingestion timestamp.
- Compute a hash over the historized attributes.
- Compare the incoming hash with the current dimension row.
- If the hash differs, close the old row (*effective_end_ts* set, *is_current* = *false*) and insert a new current row with a new surrogate key.

- Resolve each fact record to the dimension version valid at its event time (event timestamp between *effective_start_ts* and *effective_end_ts*).

9. Data Validation Rules (Bronze to Silver)

Rule	Handling
Customer email validity	Invalid rows written to customers_rejects
Country and currency standardization	Normalize codes (e.g. USA to US, UK to GB)
Latest-record deduplication	Keep latest by order_id + order_line_id, or by event_id
Quantity and key validation	Reject zero quantity, null product_id and null store_id
Inventory sanity	Reject negative on_hand_after values
Channel standardization	Map WEB, web, Online into a controlled channel vocabulary

10. Observability and Testing

Logging

- **Fabric Data Factory:** pipeline and activity monitoring for ingestion and orchestration.
- **Snowflake OPS schema:** watermark_config, pipeline_audit, activity_audit, data_quality_results.
- **Databricks:** notebook_run_log capturing run_id, step_name, input / output / rejected counts, status, error message and timestamps.

Testing

Notebooks stay thin. Business rules live in Python modules under *src/* with pytest tests under *tests/*, covering country and currency normalization, channel and loyalty-tier mapping, sales and margin formulas, inventory alert classification, SCD Type 2 change detection and duplicate removal.

11. Snowflake Design

Schema	Contents
RAW	Source-aligned batch tables
STAGING	Cleaned or intermediate exchange tables, if required
OPS	Watermarks, pipeline audit, activity audit, data-quality results
MARTS	Gold tables published for analytics

Incremental publishing uses **MERGE**. Roles follow least privilege, separating ingestion, transformation and reporting access.

12. Gold Marts and Dashboards

- **customer_analytics:** repeat-purchase behavior, loyalty tiers, customer value and counts.
- **inventory_alerts:** low-stock, out-of-stock and inventory exception analysis.
- **sales_executive_mart:** daily sales, regional and channel performance, gross margin and order KPIs.

Power BI in Fabric connects to the Snowflake MARTS schema. Planned pages: Executive Overview, Sales Performance, Customer & Loyalty, Product Performance, Inventory Alerts.

13. Databricks Job Execution Order

Notebook	Purpose
00_project_setup.py	Create schemas, control tables and configuration
10_ingest_bronze.py	Read Snowflake RAW batch data into Bronze Delta tables
20_customer_scd2.py / 21_product_scd2.py	Build customer and product SCD Type 2 dimensions
22_dimensions.py	Build store and date dimensions
23_fact_sales.py / 24_fact_inventory.py	Build validated sales and inventory-event facts
91_data_quality_checks.py	Write DQ results and rejected-record counts
30 / 31 / 32 _*.py	Build customer analytics, inventory alerts and sales executive marts
40_publish_snowflake.py	Publish Gold marts to the Snowflake reporting schema

14. Skills Demonstrated

Area	Evidence in this project
Orchestration	Fabric Data Factory pipelines, on-prem gateway, watermark-driven incremental loads
Transformation	PySpark / SQL on Databricks, Delta Lake, medallion layering
Data modeling	Star schema, conformed dimensions, SCD Type 2, surrogate keys
Data quality	Validation rules, reject tables, DQ result logging
Warehousing	Snowflake schemas, MERGE, role-based access
Engineering practice	Reusable modules, pytest suite, audit and run logging
BI delivery	Power BI dashboards on curated marts

15. Implementation Plan (10 Working Days)

Day	Deliverable
1	SQL Server source tables with dirty sample data; Snowflake databases and schemas
2	Fabric Data Factory and gateway; SQL Server to Snowflake RAW
3	JSON / Parquet loads; watermark and audit tables
4	Databricks project structure and Bronze Delta tables
5	Cleaning, deduplication, Silver dimensions and facts
6	Customer and product SCD Type 2
7	Gold marts and inventory alerts
8	Publish to Snowflake; Power BI dashboards
9	pytest suite, DQ checks, failure and reject scenarios

Day	Deliverable
10	Documentation, architecture diagram, screenshots, README

Note: the project is sized for small datasets. Databricks Free Edition and Snowflake trial credits have usage limits, so synthetic data volumes should stay modest.